

⇒ Overdetermined

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too many equations
less unknowns.

⇒ We get an approximation such that

$\|Ax - b\|^2$ is minimum
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 $\hat{x}$

Normal Equations

The minimizers of $\|Ax - b\|^2$ are the solutions of

$$A^T A x = A^T b$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

Gradients

$$\cdot \nabla_x f = \left(\frac{\partial f}{\partial x_1} \quad \dots \quad \frac{\partial f}{\partial x_n} \right)^T$$

• Show that $\nabla_x (c^T x) = c$

$$c^T x = [c_0 \ c_1 \ \dots \ c_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= c_0 \cdot x_1 + c_1 \cdot x_2 \dots c_n x_n$$

$$f = c_0 (x_1 + x_2 \dots x_n)$$

$$\nabla_x (c^T x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \cdot \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = c$$

$$\Rightarrow \nabla_x (x^T M x) = (M + M^T)x$$

$$\Rightarrow \text{Show that } \nabla_x \|Ax - b\|^2 \Rightarrow 2A^T(Ax - b) = 0$$

$$\Rightarrow \nabla_x (Ax - b)^T (Ax - b)$$

$$\Rightarrow (x^T A^T - b^T) (Ax - b)$$

$$\Rightarrow \nabla_x (x^T A^T A x - b^T A x - x^T A^T b + b^T b)$$

$$\Rightarrow 2(A^T A + (A^T A)^T)x - (A^T b) - \underbrace{A^T b}$$

$$\Rightarrow 2(A^T A + A A^T)x - 2A^T b \quad \begin{array}{l} \text{because} \\ \nabla(A^T) = (\nabla A)^T \end{array}$$

$$\Rightarrow 2A^T A x - 2A^T b = 2A^T(Ax - b)$$

$$\text{for minima } 2A^T(Ax - b) = 0$$

Exercise

① If $P = P^2 = P^T$ then P is an orthogonal projection onto $\text{Col}(P)$

$$\Rightarrow P = A (A^T A)^{-1} A^T$$

$$\rightarrow b = \underbrace{Pb}_{\text{projection}} + \underbrace{(b - Pb)}_{\text{else}}$$

$$\langle Py, b - Pb \rangle$$

$$\Rightarrow y^T P^T (b - Pb)$$

$$= y^T P (I - P) b \quad [\text{as } P^T = P]$$

$$= y^T (P - P^2) b \quad [\text{as } P^2 = P]$$

$$= 0 \quad \text{for all } y$$

② Show that the projector of Projection Theorem is idempotent & symmetric.

$$\Rightarrow \begin{array}{ll} P^2 = P & \text{so idempotent} \\ P = P^T & \text{so symmetric} \end{array}$$

Singular Value Decomposition

$A \in \mathbb{R}^{m \times n}$ then there exists U, Σ, V s.t.

$$A = U \Sigma V^T$$

U, V are orthonormals, Σ is diagonal
entries $\sigma_1 \geq \sigma_2 \dots \sigma_r > 0$

A has rank r

$\Rightarrow \sigma_i$ are called singular values.

Spectral Norm

$\|A\|_2 \rightarrow$ largest singular value.

$\Rightarrow$ The "most" a matrix can stretch a vector.

$\Rightarrow$ Show that $\|A\|_2$ is a norm

Norm should be $\Rightarrow$

$$\|A\|_2 > 0$$

$$\|A\| = 0 \text{ when } A = 0$$

$$\|\lambda x\| = |\lambda| \cdot \|x\|$$

$$\|x + y\| \leq \|x\| + \|y\|$$

$$\textcircled{1} \quad \|A\|_2 > 0$$

$$\text{since } \sigma_i = \sqrt{\lambda_i}$$

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always true

Hence max is also positive.

$$\textcircled{2} \quad \|A\|_2 = 0 \quad \text{when } A = 0$$

When $A = 0$,

$$\sigma_i = 0 \quad \forall i = \{1, \dots, r\}$$

$$\text{Hence } \max(\sigma_i) = 0$$

$$\textcircled{3} \quad \|kA\|_2 = |k| \|A\|_2.$$

$$\Rightarrow A = U \Sigma V^T$$

$$kA = kU \Sigma V^T$$

$$kA = U \underbrace{(k\Sigma)}_{\sigma_i \rightarrow |k|\sigma_i} V^T \quad \text{because } \sqrt{\lambda_i}$$

$$\max(|k|\sigma_i) = |k| \cdot \max(\sigma_i)$$

$$\textcircled{4} \quad \|A + B\|_2 \leq \|A\|_2 + \|B\|_2$$

⇒ Norm of

$$A \left\| \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \right\|_2$$

$$A^T A = \begin{bmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\|A\|_2 = 2$$

$$\Rightarrow \det \begin{vmatrix} 3-\lambda & 1 & 1 & -1 \\ 1 & 3-\lambda & -1 & 1 \\ 1 & -1 & 3-\lambda & 1 \\ -1 & 1 & 1 & 3-\lambda \end{vmatrix}$$

$$\Rightarrow \det \begin{vmatrix} 3-\lambda & 1 & 1 & -1 \\ 0 & 4-\lambda & 0 & 4-\lambda \\ 1 & -1 & 3-\lambda & 1 \\ 0 & 0 & 4-\lambda & 4-\lambda \end{vmatrix}$$

$$\Rightarrow \det \begin{vmatrix} 4-\lambda & 0 & 4-\lambda & 0 \\ 0 & 4-\lambda & 0 & 4-\lambda \\ 1 & -1 & 3-\lambda & 1 \\ 0 & 0 & (4-\lambda) & 4-\lambda \end{vmatrix}$$